Exploited Tropical Pelagic Ecosystems Observatory

An information system for the collection, storage, processing and dissemination of various data (fisheries, on-board observers, VMS, surveys, etc.)

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https://github.com/OB7-IRD/training_aerme

Introduction

What is the Exploited Tropical Pelagic Ecosystems Observatory? (Ob7)

What is exactly?

- Four components
  - An “**observatory**”
  - Several “**ecosystems**”
  - “**tropical pelagic**”
  - “**exploited**”

OBSERVATORY
OF EXPLOITED
TROPICAL
PELAGIC ECOSYSTEMS

An “observatory”

- If we take a look to the definitions
  - “establishment specialising in astronomical, meteorological or volcanic studies”
  - “observation post located generally in a elevated position”
  - “a structure responsible for observing economic, political or social events”
- Ob7 is another thing !
  - It is a **set of resources** (human and material) under the umbrella of the **French National Research Institute for Sustainable Development**
    - Created in **2000** under the name of **OSIRIS** (“observatoires et systèmes d’information des pêches tropicales”)
    - Became in **2005** “**l’Observatoire Thonier**”
    - Since **2016** the **Ob7** for Exploited Tropical Pelagic Ecosystems Observatory, under the responsability of Julien Lebranchu (julien.lebranchu@ird.fr)

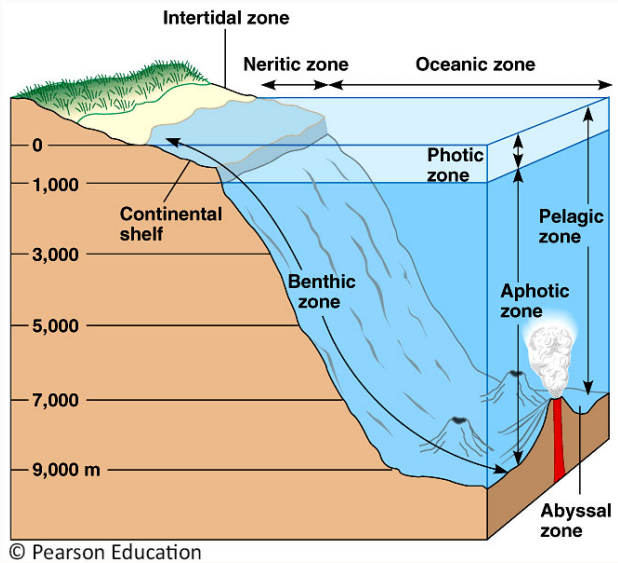
Several “ecosystems”

- What is an ecosystems ?

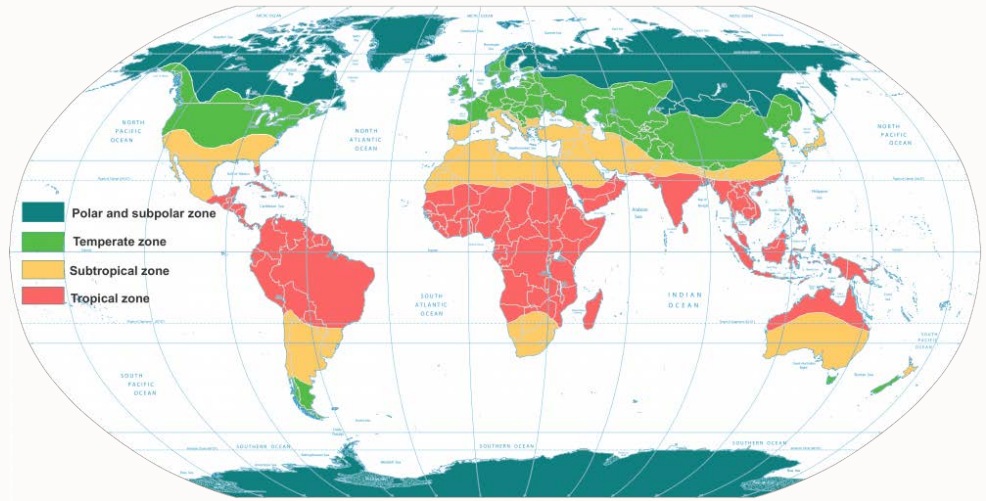
“system formed by an environment (**biotope**) and by all the species (**biocenosis**) that live, feed and reproduce there”

“Tropical pelagic”

- **Pelagic** area



- **Tropical** area



“Exploited”

- Human exploitation
- Related to the halieutic ressources = **fisheries**

To summarise

And move forward

- The Exploited Tropical Pelagic Ecosystems Observatory is a set of resources, **humans** and **materials**, under the umbrella of the French IRD
- Its aims are to **collecting**, **storing**, **processing** and **sharing** data focusing on the French tuna fisheries in the Atlantic and Indian Oceans
- Furthermore, he provides support and expertise among the **RFMO**s (**R**egional **F**isheries **M**anagement **O**rganisations) and **partners** (Spanish Institute of Oceanography, Seychelles Fishing Authority, …)

ICCAT CICTA CICAA

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INSTITUTO ESPAÑOL DE OCEANOGRAFÍA 1914

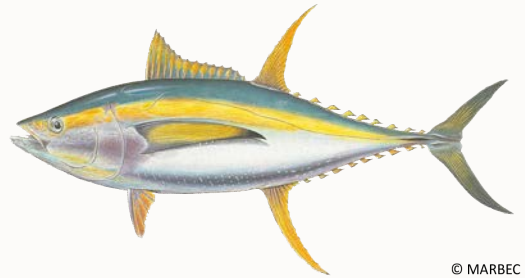
SEYCHELLES FISHING AUTHORITY

Targeted species and fishing technics

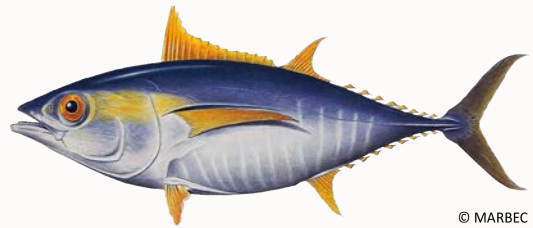
Major tunas, bycatch, floating objects and co.

Major tunas

- Yellowfin tuna (YFT, *Thunnus albacares*)

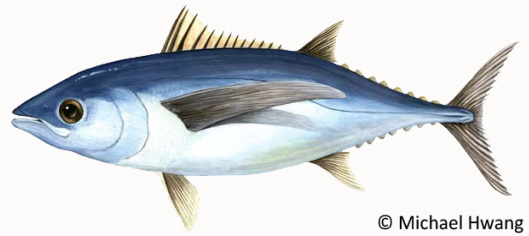


- Bigeye tuna (BET, *Thunnus obesus*)



- Skipjack tuna (SKJ, *Katsuwonus pelamis*)

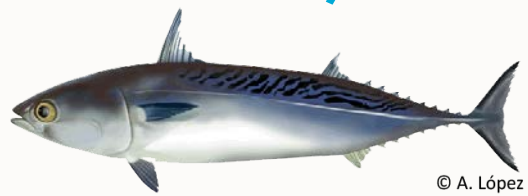
- Albacore tuna (ALB, *Thunnus alalunga*)



Bycatch species part 1

Minor tunas (neritic and thazards)

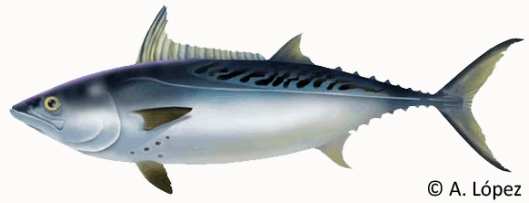
- bullet tuna (FRI, *auxis rochei*)



- king mackerel (KGM, *scomberomorus cavalla*)

- atlantic bonito (BON, *sarda sarda*)

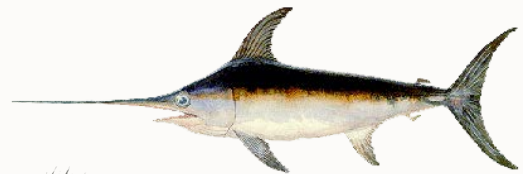
- frigate tuna (LTA, *auxis thazard*)



Bycatch species part 2

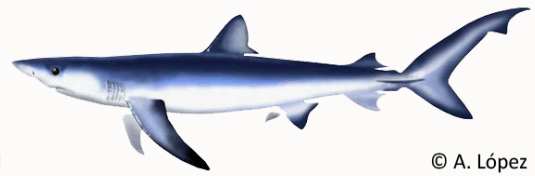
Billfish and pelagic sharks species

- swordfish (SWO, *xiphias gladius*)

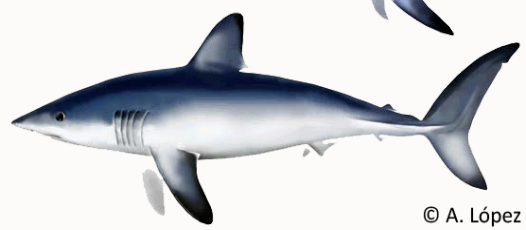


- sailfish (SAI, *istiophorus albicans*)

- blue shark (BSH, *prionace glauca*)



- portbeagle shark (POR, *lamna nasus*)



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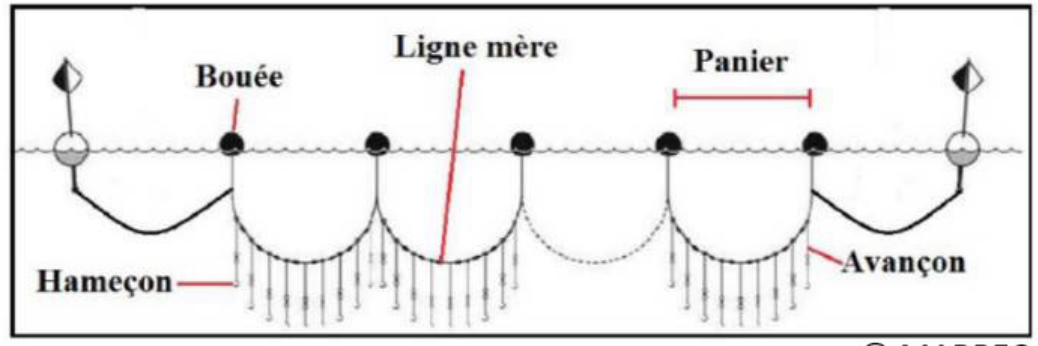
Three fishing technics part 1

Bait boat

Three fishing technics part 2

Longlines



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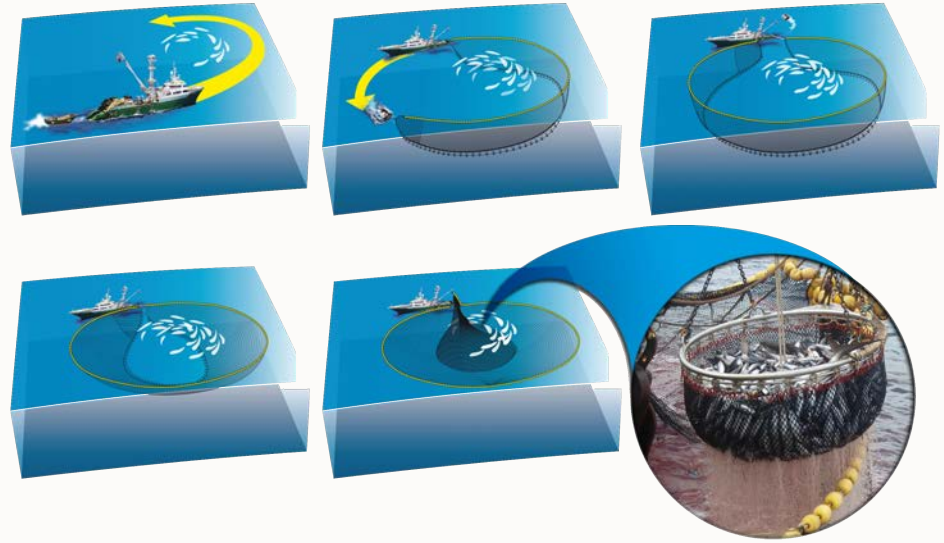
Three fishing technics part 3

Purse seiner



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Two fishing schools

Free fishing school or floating object

- FOBs: **F**loating **OB**jects
  - **natural** or **artificial**

Observatory’s aims

Why do we need this kind of structure?

Why we need an observatory?

Collection, storage, treatment and sharing of data

- Each country exploiting fisheries resources must declare its activities
  - What does that mean?
    - **Transversal variables**: effort, capacity and capture
  - To whom and why?
    - **European Commission** (if EU member, used for example regarding fisheries agreements) and **RFMOs** (population dynamics)
- **RFMOs** (**R**egional **F**isheries **M**anagement **O**rganisations): area and associated species
- These organisations are composed of the **coastal countries** of the region concerned and **countries with fisheries interests** in the region
- **Advisory** capacity, but also capacity to **define** technical measures and obligations
  - **Recommendations** and **resolutions**

RFMOs areas of interest?

For large pelagics, 5 RFMOs

The International Commission for the Conservation of Atlantic Tunas

The Indian Ocean Tuna Commission

Inter-American Tropical Tuna Commission

The Western and Central Pacific Fisheries Commission

Rotation around 4 axes

Collection, storage, treatment and sharing of data

- **Collection**: recovery of declarative data (logbook for example), sampling, fleet characteristics, etc.
- **Storage**: management in databases and administration of these databases
- **Treatment**: processes of checking/correcting data to ensure quality data with the least possible bias or specific exploitation
- **Sharing**: expertise and coordination at several levels, for research projects, in support of stock assessments, data delivery, etc.

One specific characteristic

Focus on the European Union (EU) in these processes

- Normal each entity negotiates **on its side** and is responsible for providing data
- For the EU member states it **coordinates** all these tasks
  - Intermediary with **RFMOs**
  - For France: **DGAMPA** ("Direction générale des affaires maritimes, de la pêche et de l'aquaculture") then **DGMARE** (European Commission Directorate-General Maritime Affairs and Fisheries) then **RFMOs**
- The EU guarantees the monitoring of the fishing quota(s) by European fishermen
- Gives you more **impact** on the international scene

MARITIME AFFAIRS AND FISHERIES

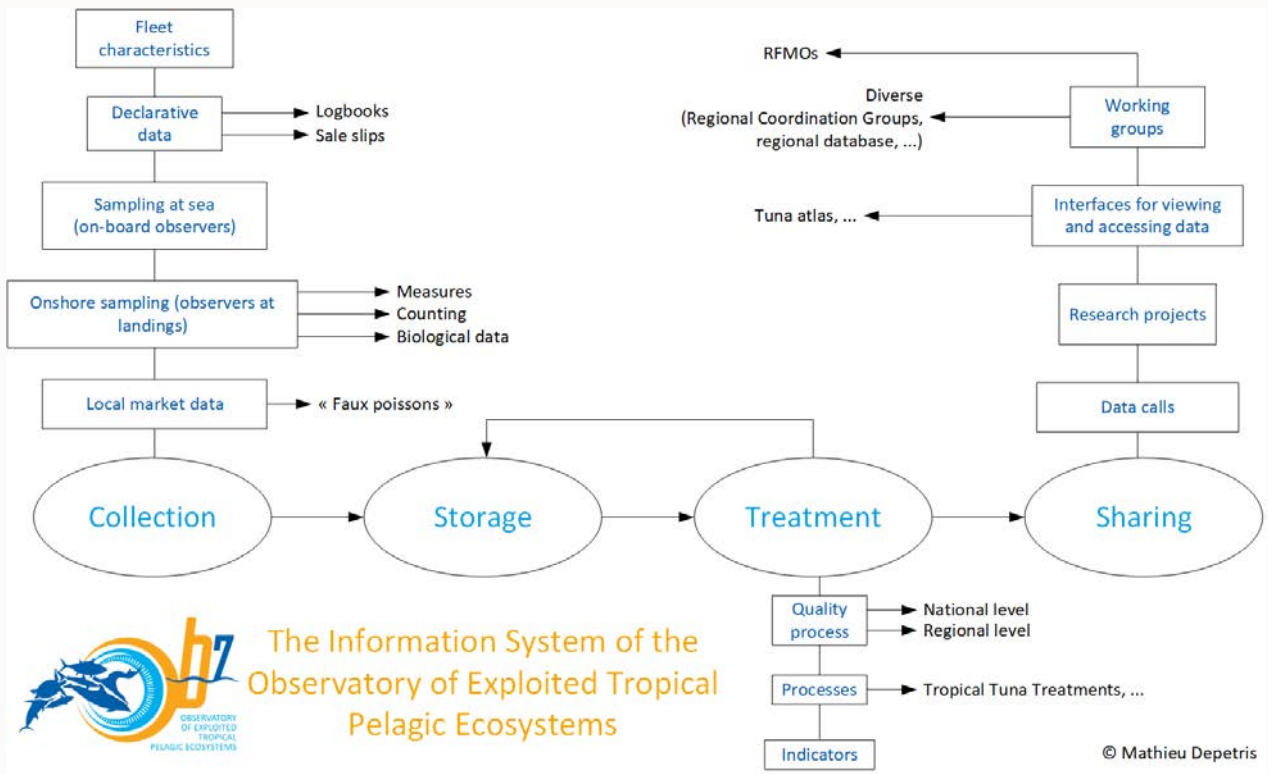
Data Collection Framework

Ob7 design

The Information System, the heart of the observatory

The heart of the observatory

The Information System: all human and material resources



https://github.com/OB7-IRD/training_aerme

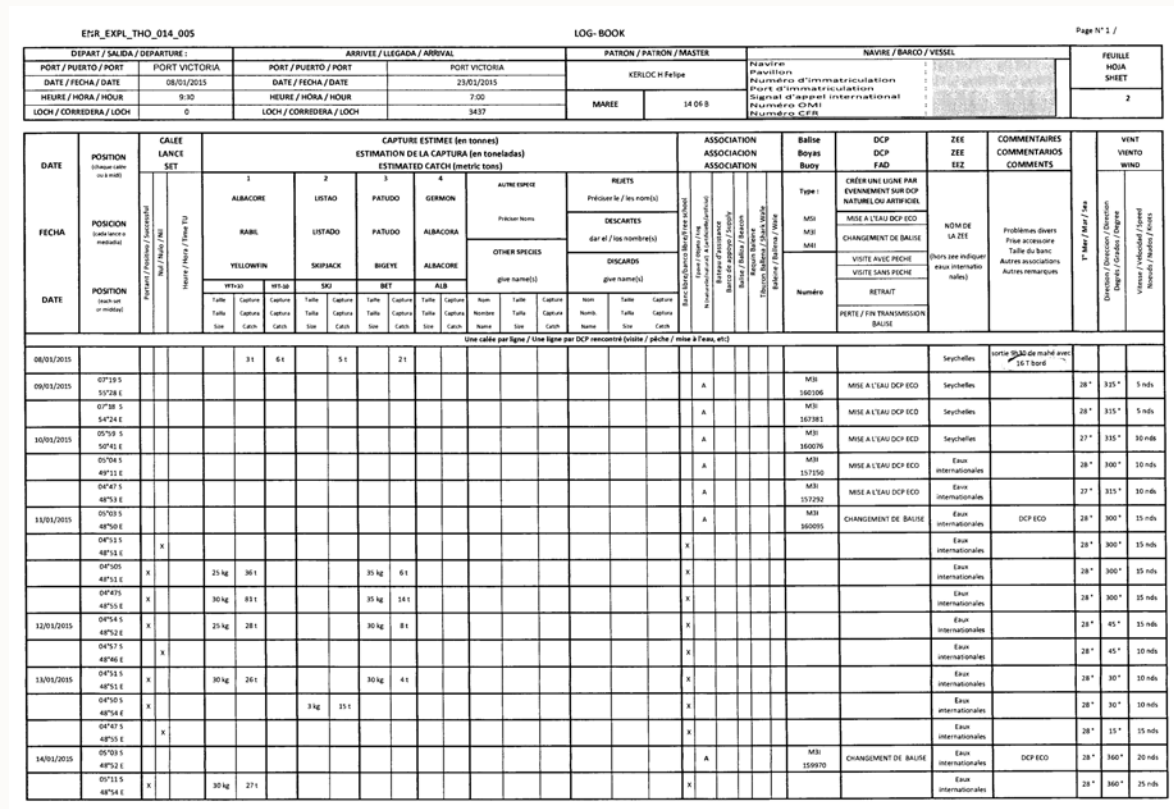
Data collection

Fleet characteristics, declarative data, onboard and onshore sampling

Data collection part 1

Fleet characteristics and declarative data

- Fleet **characteristics**: type of boat, size, active boat, etc.
- **Declarative** data
  - Logbooks and sale slips



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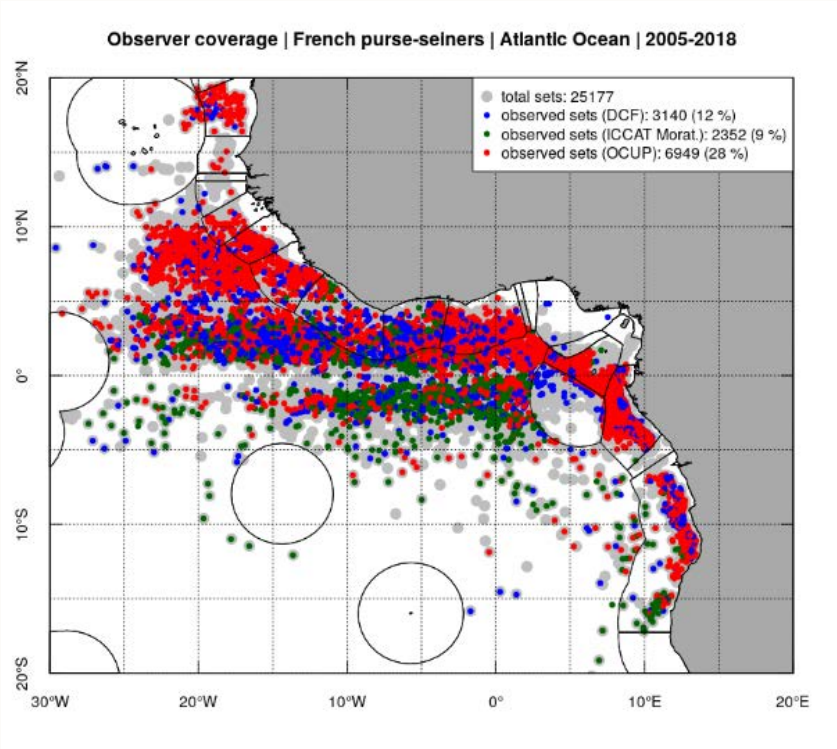
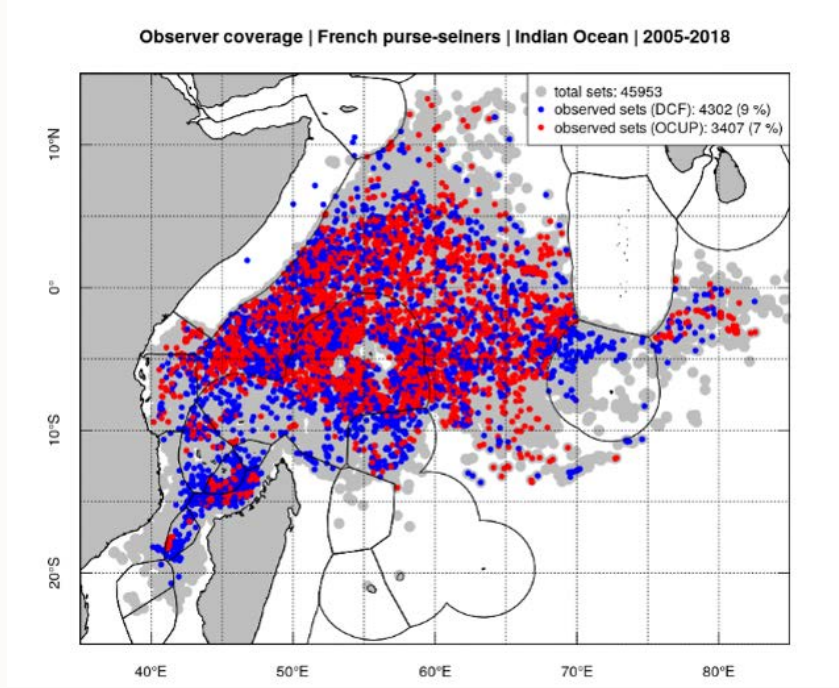
https://github.com/OB7-IRD/training_aerme

Data collection part 2

Onboard sampling

- Sampling at sea: **on-board observers**
- Priority collection of information on **discards** and **bycatch species**





https://github.com/OB7-IRD/training_aerme

Data collection part 3

Example of bycatch

https://github.com/OB7-IRD/training_aerme

Data collection part 4

Onshore sampling

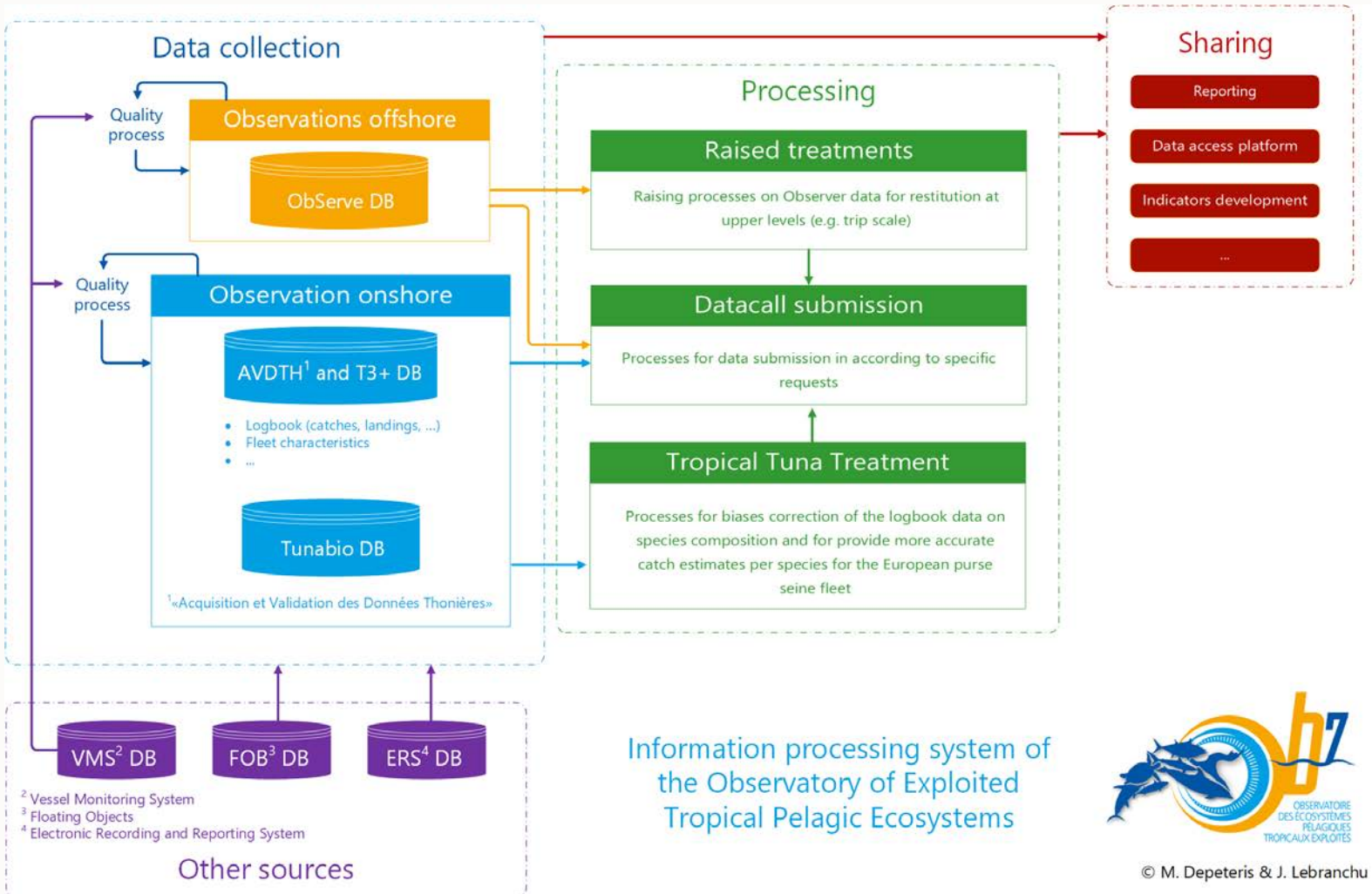
- Onshore sampling: observers at ports during **landings**
- Sampling plan based on the **well plan**
  - A catch (= **set**) is stored in one or more wells
- Individuals **measurement** and **counting**
- Collection of **biological data**: stomach contents, otolith, gonads, etc.

https://github.com/OB7-IRD/training_aerme

Ob7's Information System

Storage and treatment

The Information System



https://github.com/OB7-IRD/training_aerme

Data administration

- Administration of a **database management system**
  - Design and programming operations for the collection, storage, processing and restitution
- Ensure data quality, **identify** and **correct** errors
- Ensure data **security** and **integrity**
- Ensure that data confidentiality is **respected**

Data quality

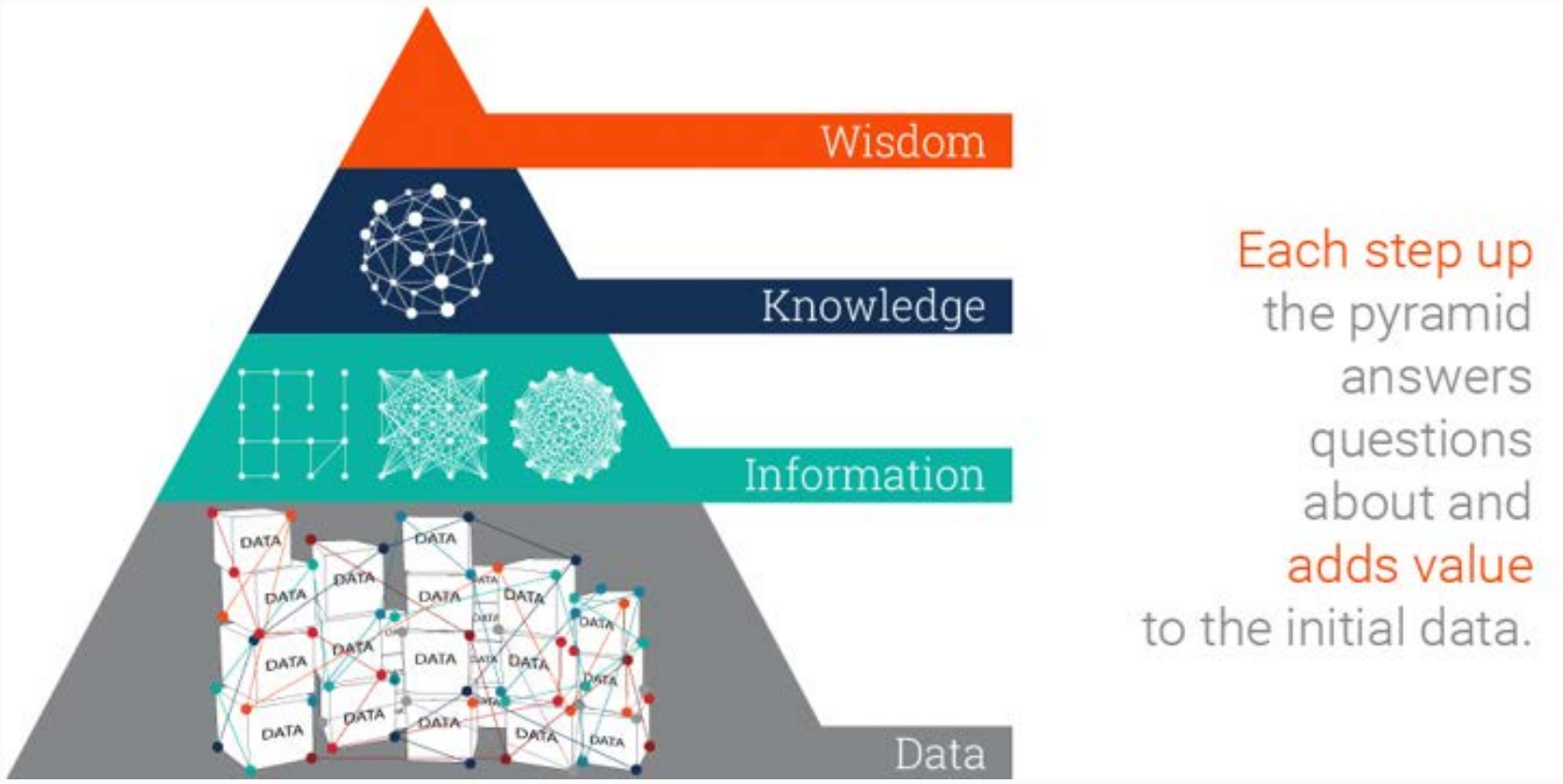
A lot of constraints

- **During** the database design
  - Referential Integrity Constraints: ensuring that data is consistent with a reference list
  - Unity constraint: to avoid duplication
  - Type constraints: integer, real,…
- **Pre-checks**
  - Check the data before integration in the database (example allow catch declarations only if a fishing trip has been successfully completed)
- **Post-checks**
  - Checking the data after entry
  - Limit the length of the sample sizes to realistic values
  - Check the consistency of the fishing season: missing days, same end date between the different sources

Data security

- **Safety** aims to protect against catastrophic failures
- **Confidentiality security** corresponds to the prevention of unauthorised access or manipulation of information and concerns the fight against intentional faults (viruses, Trojan horses, etc.)
- For the **operating system**
  - Fault-tolerant disks: RAID, Redundant Array of Independent Disks, secure data in real time
  - Backup to external media in another location
- For **Database**
  - Username/password
  - Definition of read/write access rights
- For **Network**
  - VPN (Virtual Private Network)
  - IP-based access control. Encryption of network exchanges

Data, Information, Knowledge, Wisdom (DIKW) Pyramid



https://github.com/OB7-IRD/training_aerme

Data sharing

Data calls, expertise, coordination, projects and working groups

Data deliveries

Data calls

- For diverse projects (research, internships …)
- For the **DGAMPA** (Direction générale des affaires maritimes, de la pêche et de l'aquaculture)
  - For **France** (National Work Plan …)
  - For the European Union (**F**isheries **D**ependent **I**nformation, …)
- For **RFMOs**
  - Delivery of **transversal variables** (June-July for year n-1 data)
  - Specific formats

Expertise

RFMO working groups

- The **I**nternational **C**ommission for the **C**onservation of **A**tlantic **T**unas
- The **I**ndian **O**cean **T**una **C**ommission
- To simplify, **3 levels of actions**:
  - The Working Groups
  - The Scientific Committee
  - The Commission

Coordination

At EU scale

- **R**egional **C**oordination **G**roups
  - **Main hub** for regional coordination and cooperation within the different regions contributing to the fisheries Data Collection Framework
  - Consist of experts appointed by Member States, including MS' national correspondents for data collection, and the EU Commission.
  - **Continuous** process
  - Ob7 are involve in the RCG Large Pelagics (https://www.fisheries-rcg.eu/rcg-lp/)

Regional Coordination Group
Large Pelagics

- The aim is to move from a **"national"** to a **"regional"** level.

Research support

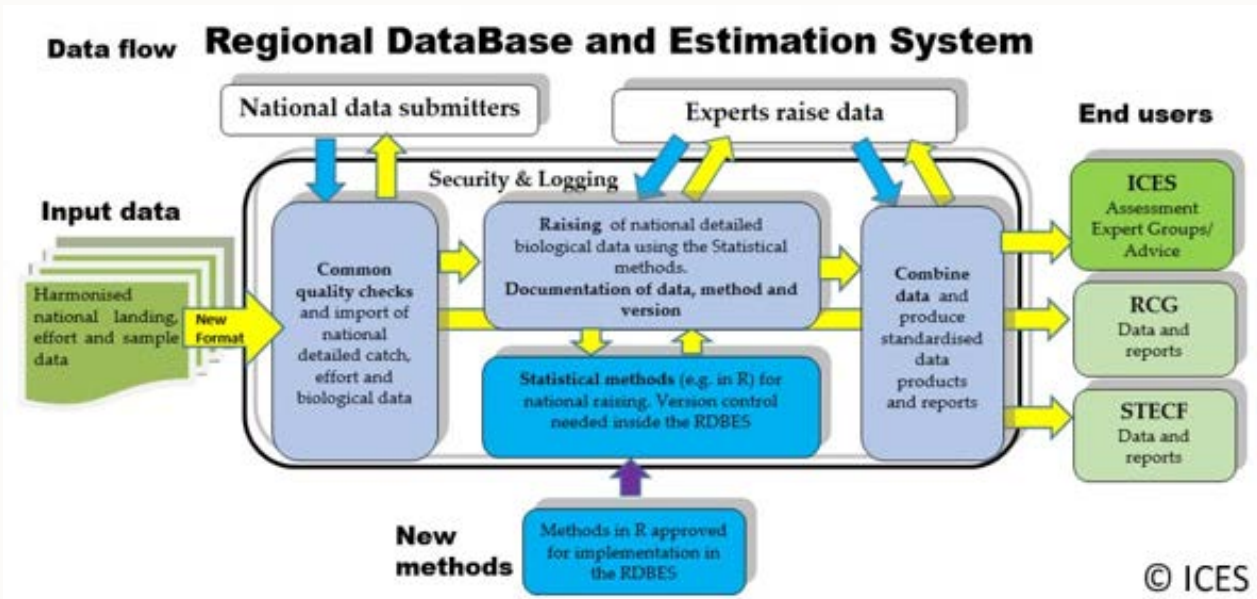
Projects

- Project FISHN'CO: Strengthening EU-MAP data collection by developing Regional Work Plans for the Regional Coordination Groups (RCG) NA NS & EA, Baltic and Large Pelagics and Economics Issues. https://www.fisheries-rcg.eu/fishnco/
- Project SECWEB: Developing mechanisms to support the planning and execution of administrative tasks and the branding and online visibility of the Regional Coordination Groups (RCGs), with the aim to establish a long-term supportive structure (SECWEB). https://www.fisheries-rcg.eu/secweb/

Other topics

Development of a regional database for large pelagics

- **R**egional **D**atabase & **E**stimation **S**ystem (RDBES, https://sboxrdbes.ices.dk/)
- Regional database for the Mediterranean and Black Seas (**Med&BS RDBFIS**, https://medbsrdb.eu/)



https://github.com/OB7-IRD/training_aerme

A case study

The onshore data collection

Background

Why we need to collect onshore data?

- **Need of data**
  - for compliance (IOTC Res. 15/01 and 15/02)
    - yearly catch data by species
    - Gridded catch data at a resolution 1° by month
  - for scientific purposes (stock assessment)
    - Catches (fishing effort)
    - Biological parameters (biometry, reproduction status, etc.)
- **Provide quality data**
  - Data independence: limiting conflict of interest between fishing business and regulation
  - Minimize biases: crew estimations reported in logbook are biased

Sampling schema

Description and design

- Data collection onshore **began** in the 60's and was **standardized** in the 90's
- **Objectives**: Estimation catch characteristics of the purse seine fisheries
  - Nominal catch by species
  - Catch and effort: catch by species by 1° square/month
  - Size distribution of the catch
- **Target population**: major tropical tuna exploited stock by purse and and baitboat fisheries
- **Region**: Atlantic and Indian ocean
- **Sample area**: ICCAT and IOTC fishing ground
- Only **main ports** are covered (Abidjan, Dakara and Victoria)
- **Primary sample unit**: landing

Sampling constraints

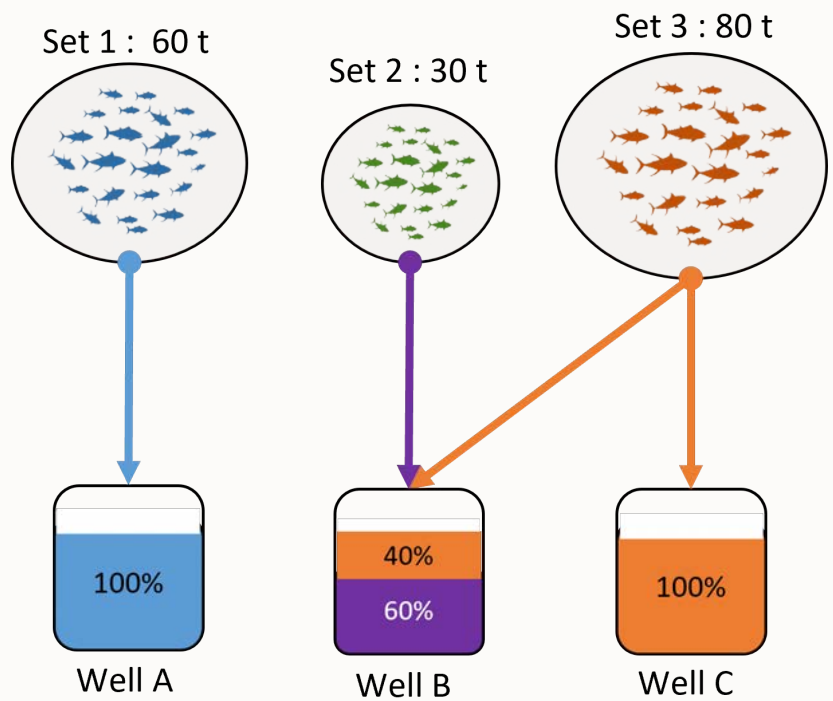
Onshore VS onboard

Key points for the collect	Onshore	Onboard
Human cost	Sample team at port	Observers onboard
Collect location	Few number of landing ports	Numerous vessels
Sampling period	At landing (3-5 days)	At each catch (every day during the trip)
Sampling point	Well sampling at landing	Sampling during the brailing
Sampling easiness	Medium (well selection)	Hard (fast loading into wells)
Other program link/conflict	-	Bycatch survey

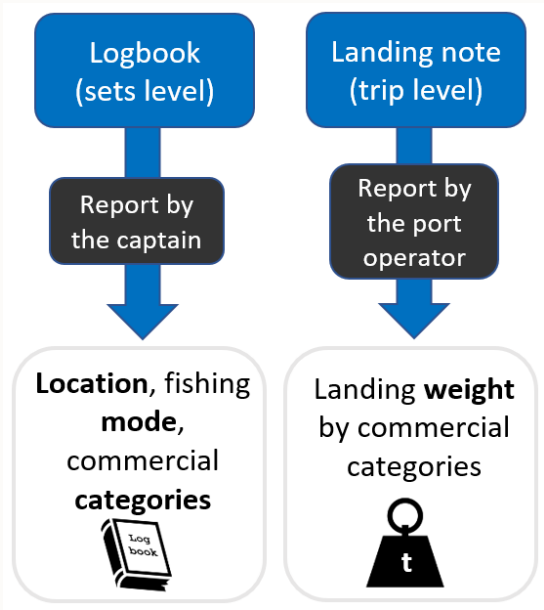
Purse seine fisheries' specificities

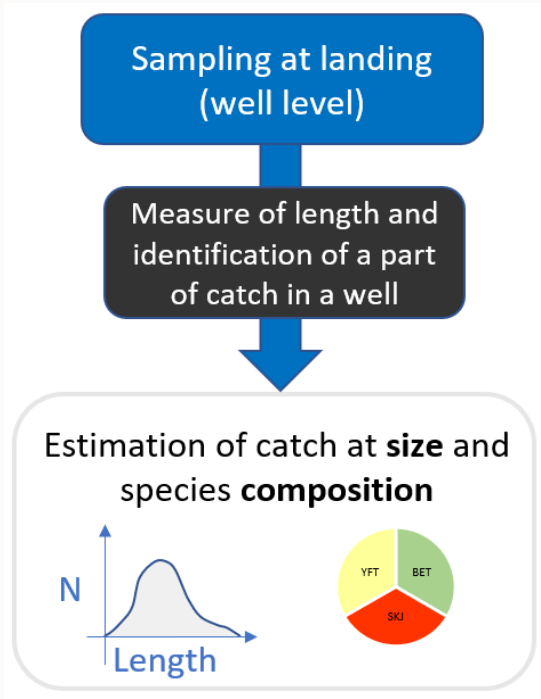
- **Multispecific catches**: major tunas, minor tunas and bycatch
- Large amount of **non-sorted catch**
- **Misclassification** among species: similarity between YFT and BET juveniles
- Mix of **several sets in wells**

A sample = a mix of species and sets in a well



Data collection





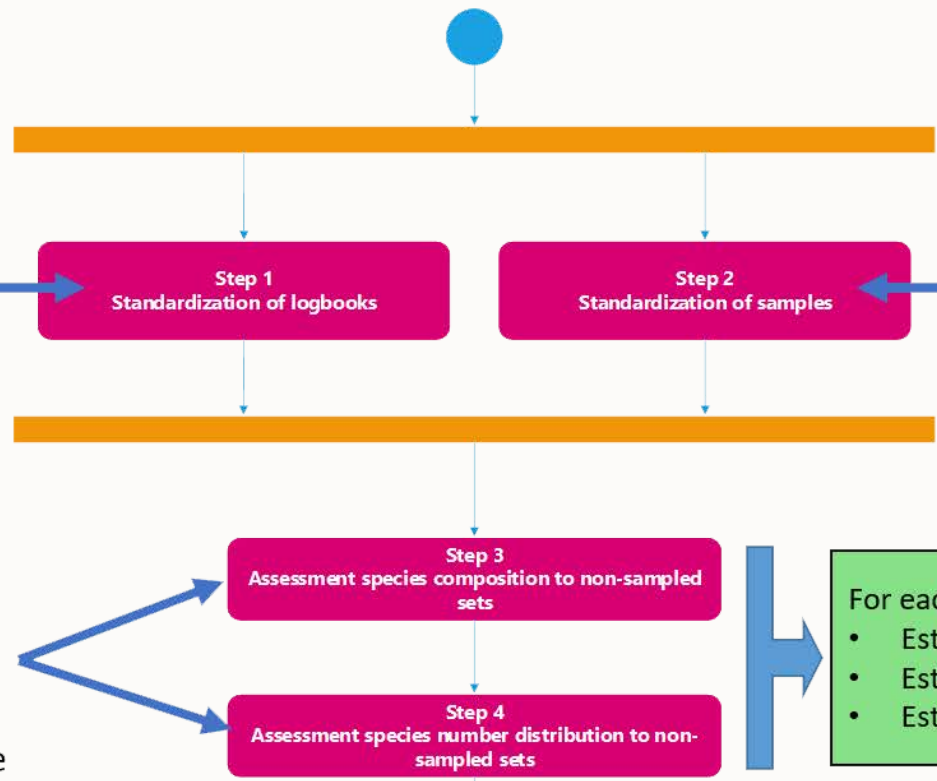
Protocole principle

- Wells selected with quality criteria (homogeneity in fishing mode and composition spatio-temporal limitation)
- Identify: 500 fishes for log school and 200 fishes for free school
- Length measure: Number vary with species composition

Data processing

Tropical Tuna Treament (T3)

- **Adjust** catches to landing weight
- **Standardize** weight categories



- **Standardize** measurement
- **Allocate** the sample species composition to the set level
- Use of the Weight-Length relationship (WLR)

- Define **homogenous strata**
- **Assess the species composition** and size distribution

For each species
- Estimated total catch
- Estimated Catch and Effort
- Estimated Catch at size

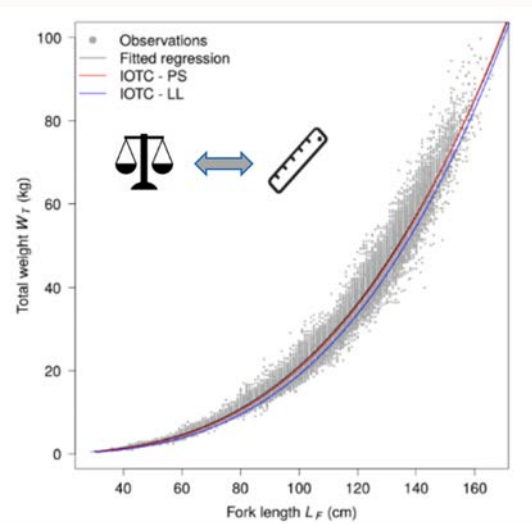
National Fishing Administration

https://github.com/OB7-IRD/training_aerme

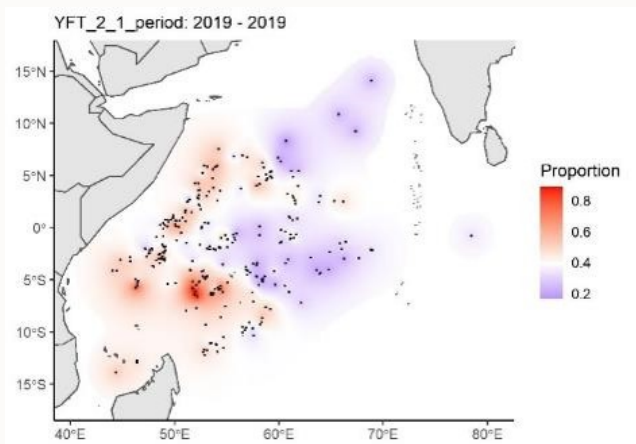
Tropical Tuna Treament part 1

Outputs examples

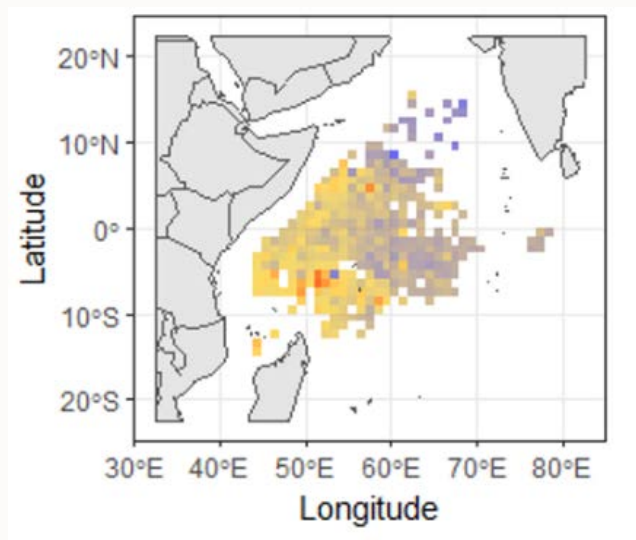
Length-weight relationship



Observed values



Fitted values

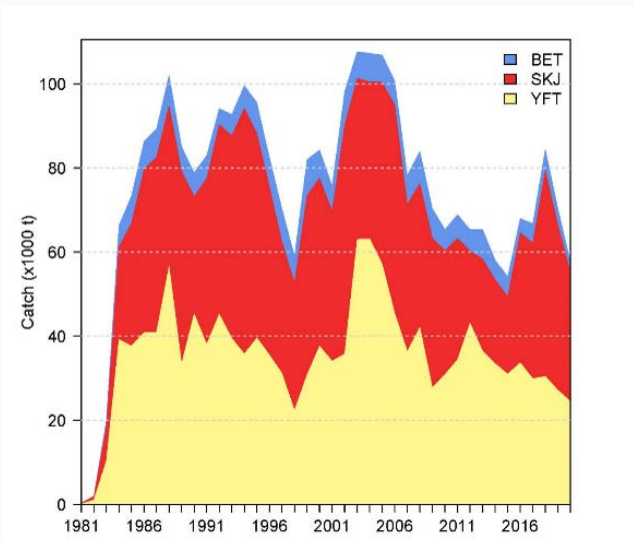


https://github.com/OB7-IRD/training_aerme

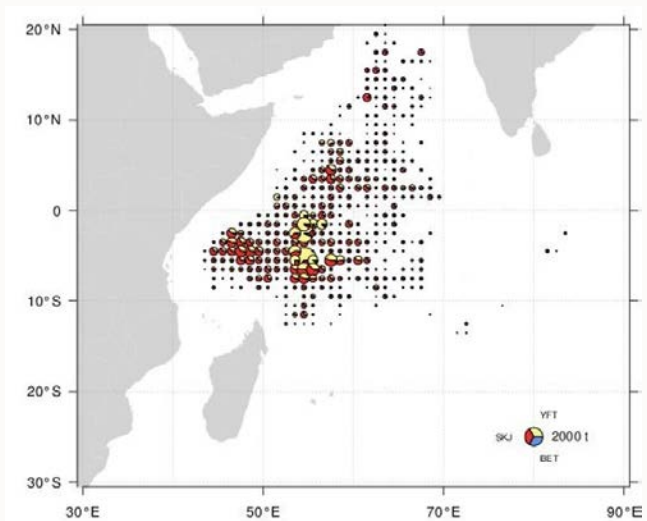
Tropical Tuna Treament part 2

Outputs examples

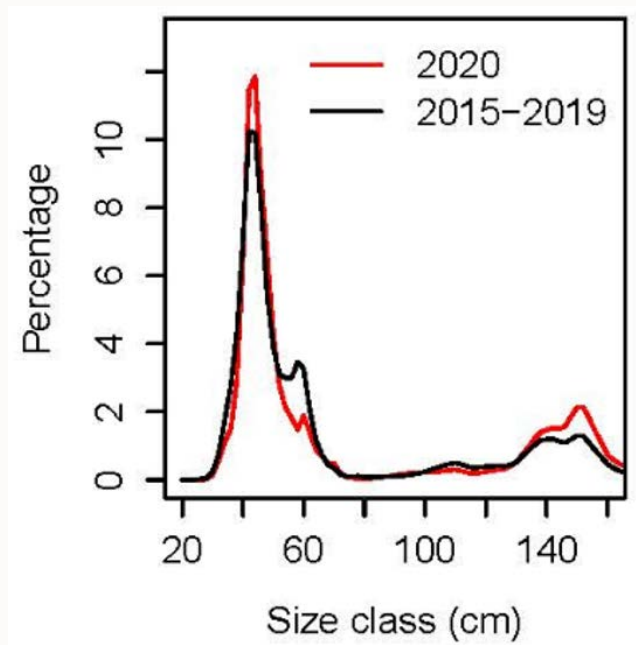
Estimated total catch



Estimated Catch and Effort



Size distribution (YFT-2020)

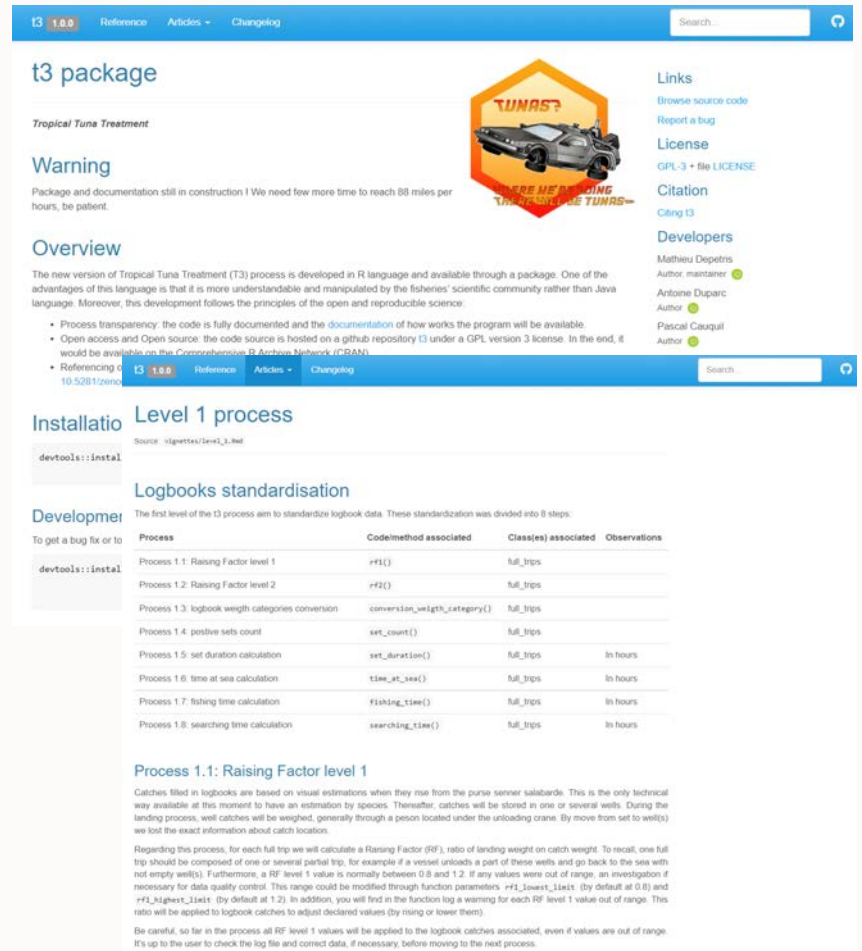


https://github.com/OB7-IRD/training_aerme

Tropical Tuna Treament part 3

Development processing

- **T3 r package**
  - **Transparency process**: code fully documented associated to documentations
    - https://ob7-ird.github.io/t3/
  - **Open access and source**: code hosted on a github repository under GPL 3 version licence
    - https://github.com/OB7-IRD/t3
    - Referencing of the software: enable result reproducibility and versioning version 1.0.0 - DOI: 10.5281/zenodo.3878125
    - Easier access for development to every group member



https://github.com/OB7-IRD/training_aerme

Thank you for your attention, do you have any questions?

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https://github.com/OB7-IRD/training_aeme

Jeu de rôle

Simulation d’une réunion du RCG LP

- Normalement réunion en anglais, la pour simplifier on passe en français
- Contenu et aide disponible dans le repertoire “role_play_rcg” du Github de la formation
- Déroulement de la simualtion :
  - Création des groupes et distribution des fiches d’identité de chaque pays - 5min
  - Présentation du contexte, ouvertures et des éléments support pour vos discussions - 10min
  - Discussion en groupe - 15min
  - Discussions entre les participants - 30min
  - Débriefing et comparaison avec la réalité - 30min

Création des groupes et distribution des fiches d’identité de chaque pays

5min

Présentation du contexte, ouvertures et des éléments support pour vos discussions

10min

Contexte

- **RCG** : “Regional Coordination Group”, groupe de coordination régional
- **Objectifs**
  - améliorer la **coordination**, au sens large et au niveau **européen**, de le collecte de données associé aux pécheries
  - Intermédiaire entre le niveau “**national**” et “**internationnal**”
  - Passage d’une dynamique “**nationale**” à une dynamique “**regionale**”
- 6 actuellement en activité
  - RCG NANSEA : Groupe de coordination régionale pour l’Atlantique Nord, la mer du Nord et l’Arctique oriental.
  - RCG Baltic : Groupe de coordination régionale pour la Baltique.
  - RCG Med&BS : Groupe de coordination régionale pour la Méditerranée et la mer Noire.
  - RCG ECON : groupe de coordination régionale sur les questions économiques et groupe pan-régional qui s’occupent de la collecte de données socio-économiques.
  - **RCG LP** : groupe de coordination régionale pour les grands pélagiques.
  - RCG LDF : groupe de coordination régionale pour la pêche à longue distance.
- **Composé** des pays européens concernés par la thématique, des utilisateurs finaux des données et des pays tiers

Sujet

- Votre objectif
  - Trouver un consensus sur le développement d'une base de données régionale pour le RCG LP
  - Pour simplifier on oublie tous les aspects techniques liée aux bases de données
- Axe principal
  - Volonté commune d'avancer sur le sujet
  - N'oubliez pas d'être bienveillant et constructifs entre vous !

Pour vous guider

Agenda de la réunion

- Choix du système (RDBES/RDBFIS/Autre système)
- Quels investissements humain et financier pour chaque pays ?
- Plan d’action (court, moyen et long terme)